

Machine Learning Algorithms

This report explains four machine learning algorithms: Logistic Regression, Decision Tree Classifier and K-Nearest Neighbors (classification), and Linear Regression (regression).

1. Logistic Regression (Classification)

What it is

A supervised learning algorithm used to predict a category, usually two classes such as Yes/No. Despite its name, it is a classification algorithm.

How it works

- It takes a weighted sum of the input features.
- The sigmoid function converts this sum into a probability between 0 and 1: $P = 1 / (1 + e^{-z})$.
- If the probability is 0.5 or more, the output is Class 1, otherwise Class 0.

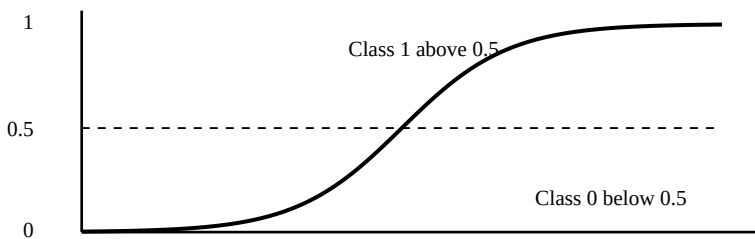


Fig 1: Sigmoid curve used in Logistic Regression

Where it is used

Medical diagnosis, spam filtering, credit scoring and customer churn prediction.

Real-world application

Email spam detection: the model predicts whether an email is spam or not spam.

Advantages

- Simple, fast and easy to understand.
- Gives probabilities, not just labels.
- Works well on small datasets.

Limitations

- Assumes a linear relationship between features and the outcome.
- Does not perform well on complex, non-linear data.

2. Decision Tree Classifier (Classification)

What it is

A supervised algorithm that makes decisions using a tree-like structure of questions.

How it works

- It picks the feature that best splits the data (using Gini or Entropy).
- Each internal node asks a question, each branch is an answer, and each leaf gives the final class.
- Splitting continues until the data is separated or a depth limit is reached.

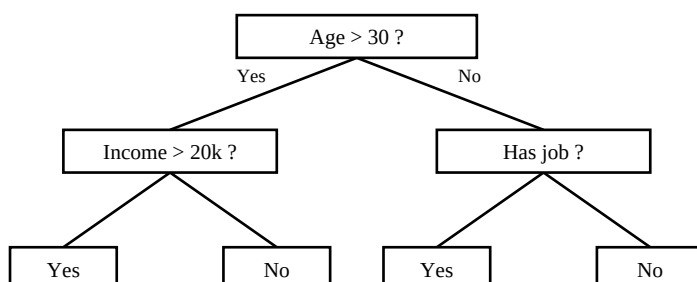


Fig 2: Simple decision tree

Where it is used

Loan approval, medical diagnosis, fraud detection and customer segmentation.

Real-world application

Banks use it to decide whether to approve a loan based on age, income and job status.

Advantages

- Easy to understand and visualize.
- Needs very little data preparation (no scaling needed).
- Handles both numeric and categorical data.

Limitations

- Can overfit the training data.
- A small change in data can change the whole tree.

3. K-Nearest Neighbors, KNN (Classification)

What it is

A simple supervised algorithm that classifies a new data point based on the classes of its closest neighbors.

How it works

- Choose a value of K (for example 3 or 5).
- Calculate the distance (usually Euclidean) from the new point to all training points.
- Select the K nearest points; the majority class among them becomes the prediction.

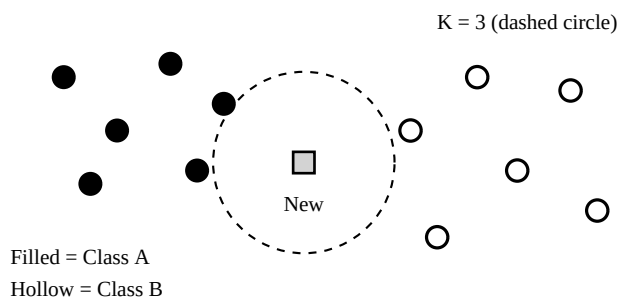


Fig 3: KNN classifying a new point using its 3 nearest neighbors

Where it is used

Recommendation systems, handwriting recognition, image classification and medical diagnosis.

Real-world application

Movie or product recommendation: suggesting items liked by similar users.

Advantages

- Very simple and easy to implement.
- No training phase is needed.
- Works well for small datasets.

Limitations

- Slow on large datasets because it calculates distance to every point.
- Needs feature scaling and a good choice of K.
- Sensitive to noisy data.

4. Linear Regression (Regression)

What it is

A supervised algorithm used to predict a continuous numeric value, such as price or marks.

How it works

- It fits a straight line through the data: $y = m x + c$, where m is the slope and c is the intercept.
- The best line is found by minimizing the squared difference between actual and predicted values (Least Squares method).

- The line is then used to predict the output for new inputs.

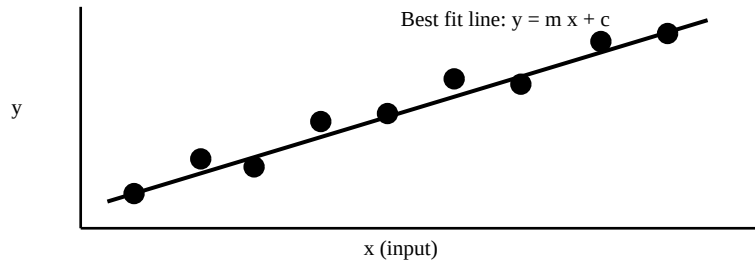


Fig 4: Linear Regression best fit line

Where it is used

House price prediction, sales forecasting, salary estimation and risk analysis.

Real-world application

Predicting the price of a house from its area, number of rooms and location.

Advantages

- Simple, fast and easy to interpret.
- Works well when the data has a linear trend.

Limitations

- Cannot capture non-linear relationships.
- Sensitive to outliers.
- Performs poorly if the input features are highly correlated.